

Version 2 *misbehavior* and banning specification.

Models the receiving side of the draft’s “Misbehavior and Banning” section: a local node scoring the peers it is connected to. The draft constrains every penalty by a provability principle — a node MUST NOT penalize behaviour that a conformant peer acting on honest but divergent state could produce — and motivates it: penalties on weaker evidence let an attacker induce honest nodes to ban one another. This module turns that sentence and its converse into checkable properties.

Remote peers are honest or *Byzantine*. Honest peers may legitimately send:

|                   |                                                                                                                                                                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| divergent headers | contiguous, valid proof of work, but not connecting to the local chain: the peer follows another fork (“Block Announcements”, “Headers-First Synchronization” — MUST NOT penalize).                                                                   |
| requested invalid | a block the local node itself requested by hash whose content fails consensus validation: the responder served exactly the bytes the hash names; blame lies with the announcer (“Misbehavior and Banning”, exemption “Content of requested objects”). |

*Byzantine* peers additionally send provable violations from the penalty table: oversize get-headers responses ( + 20), non-contiguous headers ( + 20), and an announced block with invalid proof of work ( + 100).

Three booleans select wrong readings of the draft:

|                                        |                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------|
| <i>PenalizeNonConnecting</i> = TRUE    | → divergent headers score 20 points                                            |
| <i>PenalizeRequestedInvalid</i> = TRUE | → a requested invalid block scores 100                                         |
| <i>PerConnectionScore</i> = TRUE       | → the score is per connection, reset on reconnect, instead of keyed by address |

Under the first two, *NoHonestBan* is violated: an honest peer on another fork is banned. Under the third, *PersistentAttackerBanned* is violated: a *Byzantine* peer sheds its score by reconnecting before the threshold — the draft’s stated reason for address-keyed persistent scores.

See `../documents/v2-modeling.md` for the full write-up.

EXTENDS *Naturals*, *FiniteSets*

|                                          |                                                    |
|------------------------------------------|----------------------------------------------------|
| CONSTANT <i>Peers</i>                    | remote peers of the local node                     |
| CONSTANT <i>Byzantine</i>                | subset of <i>Peers</i> sending provable violations |
| CONSTANT <i>BanThreshold</i>             | score at which a peer is disconnected and banned   |
| CONSTANT <i>PenalizeNonConnecting</i>    | TRUE penalizes divergent headers (buggy)           |
| CONSTANT <i>PenalizeRequestedInvalid</i> | TRUE penalizes requested invalid blocks (buggy)    |
| CONSTANT <i>PerConnectionScore</i>       | TRUE resets the score on reconnect (buggy)         |

*Honest*  $\triangleq$  *Peers* \ *Byzantine*

VARIABLES

|                  |                                                                              |
|------------------|------------------------------------------------------------------------------|
| <i>score</i> ,   | [ <i>Peers</i> → 0 .. <i>BanThreshold</i> ] <i>misbehavior</i> score, capped |
| <i>banned</i> ,  | [ <i>Peers</i> → BOOLEAN ] banned addresses (absorbing)                      |
| <i>connected</i> | [ <i>Peers</i> → BOOLEAN ] currently connected                               |

*vars*  $\triangleq$   $\langle$ *score*, *banned*, *connected* $\rangle$

*Min*(*a*, *b*)  $\triangleq$  IF *a* < *b* THEN *a* ELSE *b*

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*Init*  $\triangleq$

$\wedge \text{score} = [p \in \text{Peers} \mapsto 0]$   
 $\wedge \text{banned} = [p \in \text{Peers} \mapsto \text{FALSE}]$   
 $\wedge \text{connected} = [p \in \text{Peers} \mapsto \text{TRUE}]$

The local node adds *pts* to *p*'s score; at the threshold it closes the connection with *MISBEHAVIOR* and bans the address (“Misbehavior and Banning”: score thresholds and banning).

*Penalize*(*p*, *pts*)  $\triangleq$

LET *s*  $\triangleq \text{Min}(\text{score}[p] + \text{pts}, \text{BanThreshold})$   
 IN  $\wedge \text{score}' = [\text{score} \text{ EXCEPT } ![p] = s]$   
 $\wedge \text{banned}' = [\text{banned} \text{ EXCEPT } ![p] = s \geq \text{BanThreshold}]$   
 $\wedge \text{connected}' = [\text{connected} \text{ EXCEPT } ![p] = @ \wedge s < \text{BanThreshold}]$

*NoPenalty*  $\triangleq \text{UNCHANGED } \langle \text{score}, \text{banned}, \text{connected} \rangle$

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An honest peer on a divergent fork answers get-headers, or announces a header, that does not connect to the local chain. The draft forbids a penalty; *PenalizeNonConnecting* scores it anyway.

*HonestDivergentHeaders*  $\triangleq$

$\exists p \in \text{Honest} :$   
 $\wedge \text{connected}[p]$   
 $\wedge \text{IF } \text{PenalizeNonConnecting} \text{ THEN } \text{Penalize}(p, 20) \text{ ELSE } \text{NoPenalty}$

An honest peer serves a block the local node requested by hash whose content fails consensus validation — exactly the bytes the hash names (for example an artifact held for checkpointed sync). The draft exempts it; *PenalizeRequestedInvalid* scores it anyway.

*HonestRequestedInvalid*  $\triangleq$

$\exists p \in \text{Honest} :$   
 $\wedge \text{connected}[p]$   
 $\wedge \text{IF } \text{PenalizeRequestedInvalid} \text{ THEN } \text{Penalize}(p, 100) \text{ ELSE } \text{NoPenalty}$

Provable violations from the penalty table: only *Byzantine* peers produce them, and they always score.

*ByzOversizeHeaders*  $\triangleq$

$\exists p \in \text{Byzantine} : \text{connected}[p] \wedge \text{Penalize}(p, 20)$

*ByzNonContiguousHeaders*  $\triangleq$

$\exists p \in \text{Byzantine} : \text{connected}[p] \wedge \text{Penalize}(p, 20)$

*ByzInvalidPowBlock*  $\triangleq$

$\exists p \in \text{Byzantine} : \text{connected}[p] \wedge \text{Penalize}(p, 100)$

A peer may disconnect at will — in particular a *Byzantine* peer trying to shed its score.

$Disconnect \triangleq$

$\exists p \in Peers :$   
 $\wedge connected[p]$   
 $\wedge connected' = [connected \text{ EXCEPT } ![p] = FALSE]$   
 $\wedge UNCHANGED \langle score, banned \rangle$

A disconnected, unbanned peer of  $S$  reconnects. Whether its score survives the new connection is the *PerConnectionScore* switch; the draft wants scores keyed by address precisely so that reconnecting sheds nothing.

$ReconnectFrom(S) \triangleq$

$\exists p \in S :$   
 $\wedge \neg connected[p]$   
 $\wedge \neg banned[p]$   
 $\wedge connected' = [connected \text{ EXCEPT } ![p] = TRUE]$   
 $\wedge score' = [score \text{ EXCEPT } ![p] = \text{IF } PerConnectionScore \text{ THEN } 0 \text{ ELSE } @]$   
 $\wedge UNCHANGED \langle banned \rangle$

$Reconnect \triangleq ReconnectFrom(Peers)$

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$Next \triangleq$

$\vee HonestDivergentHeaders$   
 $\vee HonestRequestedInvalid$   
 $\vee ByzOversizeHeaders$   
 $\vee ByzNonContiguousHeaders$   
 $\vee ByzInvalidPowBlock$   
 $\vee Disconnect$   
 $\vee Reconnect$

Fairness encodes the hypothesis of *PersistentAttackerBanned*: a *Byzantine* peer that KEEPS misbehaving — and keeps reconnecting when disconnected — is the one that must end up banned. Fairness is on the *Byzantine* peer's own reconnection (an honest peer reconnecting must not discharge it); honest events and *Disconnect* are never forced. The configurations use a single *Byzantine* peer, so the existential fairness cannot starve one attacker in favour of another.

$Spec \triangleq$

$Init$   
 $\wedge \Box [Next]_{vars}$   
 $\wedge SF_{vars}(ByzOversizeHeaders)$   
 $\wedge SF_{vars}(ReconnectFrom(Byzantine))$

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Liveness: a persistently misbehaving *Byzantine* peer is eventually banned.

Violated under *PerConnectionScore*: reconnecting before the threshold sheds the score forever.

$$PersistentAttackerBanned \triangleq \Diamond \forall p \in Byzantine : banned[p]$$

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Safety invariants.

$$\begin{aligned} TypeOK &\triangleq \\ &\wedge \quad score \in [Peers \rightarrow 0 .. BanThreshold] \\ &\wedge \quad banned \in [Peers \rightarrow \text{BOOLEAN}] \\ &\wedge \quad connected \in [Peers \rightarrow \text{BOOLEAN}] \end{aligned}$$

The provability principle, as an outcome: no honest peer is ever banned, however divergent its chain view. Violated under either buggy penalty.

$$\begin{aligned} NoHonestBan &\triangleq \\ &\forall p \in Honest : \neg banned[p] \wedge score[p] < BanThreshold \end{aligned}$$

A banned peer is disconnected and stays banned.

$$\begin{aligned} BanIsFinal &\triangleq \\ &\forall p \in Peers : banned[p] \Rightarrow \neg connected[p] \end{aligned}$$


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